

# The Work a Verifier Needs

Certified Int8 Output Heads That Match a Production Kernel's Rounding

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Preprint · Qwen3.5-4B in SGLang on one NVIDIA GH200

[github.com/SamMausberg/verified-progress](https://github.com/SamMausberg/verified-progress)

AI (Claude Opus 5.5, GPT-6 Astra Pro) was used in developing the paper and the code.

# A verifier needs one token, yet the engine scores 248,320

Speculative decoding: a drafter proposes tokens; the target checks them all in one forward pass.

A greedy verifier accepts a drafted token only if it is the target's top-scoring entry.

**How little of the head can an exact shortcut compute?**



one logit per vocabulary entry:  
248,320 per position

# Measured on recorded engine states, then served

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## Model and engine

Qwen3.5-4B in BF16  
MTP-4B, DFlash-4B drafters  
SGLang bd66ce343e  
one GH200

## Decisions: replayed

320 public prompts, greedy  
head inputs captured  
replayed in FP64  
and in our GPU kernel  
checked inside SGLang

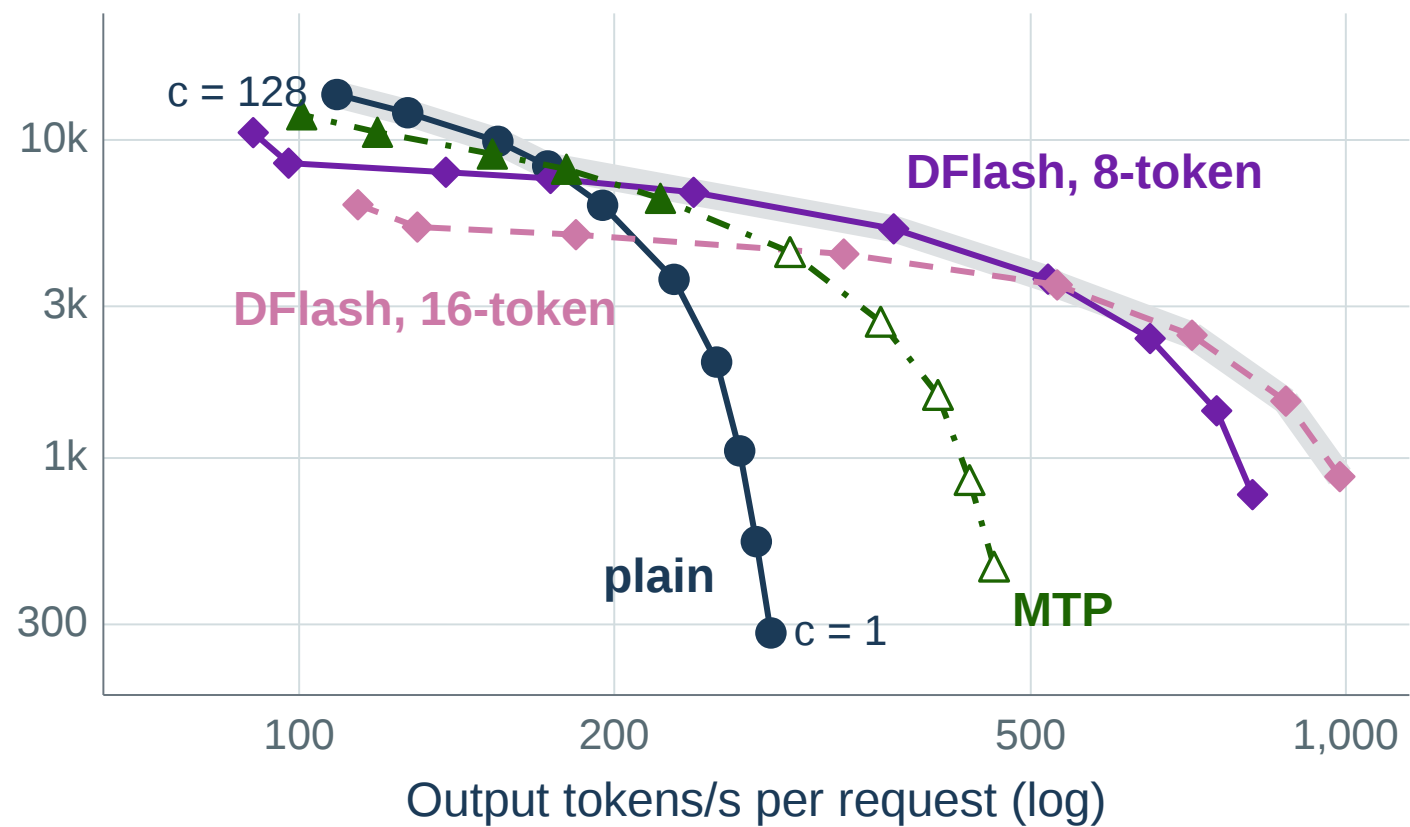
## Speed: served sweeps

fresh launch per session  
CUDA graphs, overlap checked  
 $c = 1\text{--}128$  requests in flight  
1,152 prompts  $\times$  512 tokens  
three sessions per result

**An arm** is one tuned serving configuration; **c** is the number of requests a closed-loop client keeps in flight.

# Each drafter's best arm beats plain decoding up to $c = 32$

Output tokens/s on the GPU (log)



3.06×

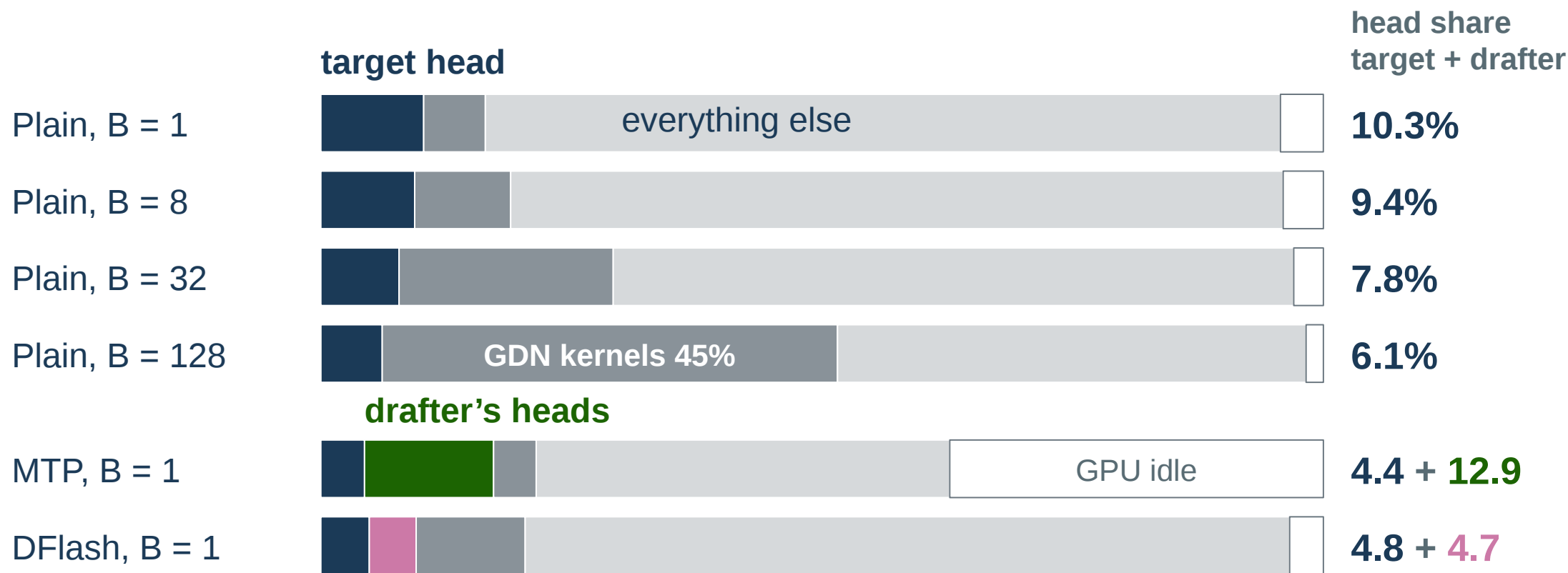
throughput at  $c = 1$ : 16-token  
DFlash over plain decoding,  
same attention

Mean accepted tokens per verify  
cycle

|           |           |
|-----------|-----------|
| DFlash-16 | 5.59–5.77 |
| DFlash-8  | 4.66–4.81 |
| MTP       | 3.24–3.28 |

From  $c = 48$  plain decoding  
leads (default admission)

# The head is 10% of a plain step at batch 1, 6% at 128

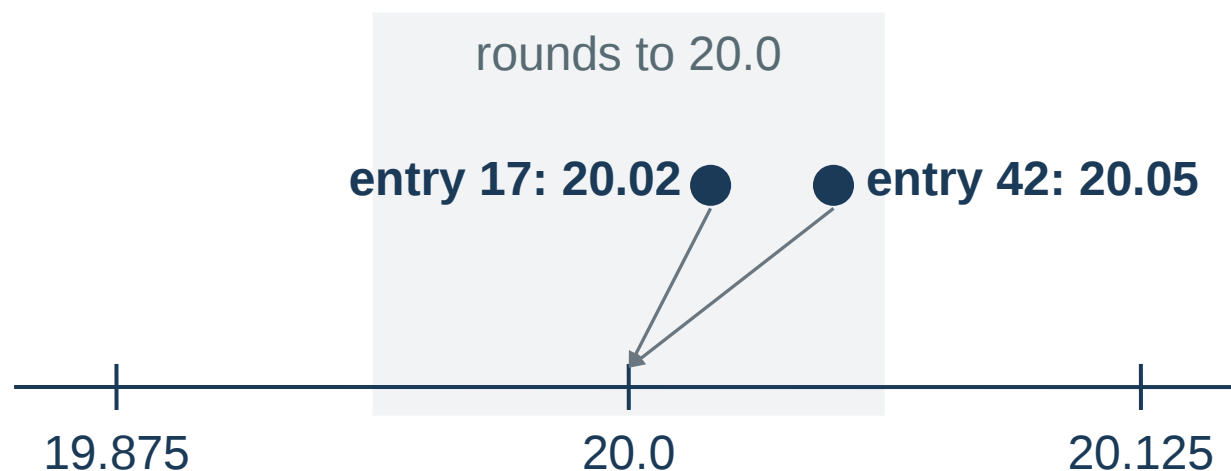


GDN: Gated DeltaNet, the linear attention in three of the model's four layers, with a recurrent state per request

**A free head would make a batch-1 plain step only about 11% faster.**

# “Exact” must mean the engine’s token, not the exact winner

A made-up example



**Exact winner: 42.** The engine returns 17: both round to 20.0, and argmax takes the first index.

## 0.5%

of greedy positions: the exact winner is not the engine’s token

## 1.0%

the top two logits tie in BF16

# Even stock SGLang changes its output with the batch

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## 167 of 320

prompts' greedy outputs differ between 1 and 32 requests in flight on one stock server

First differences come from batch-dependent kernels upstream of the head

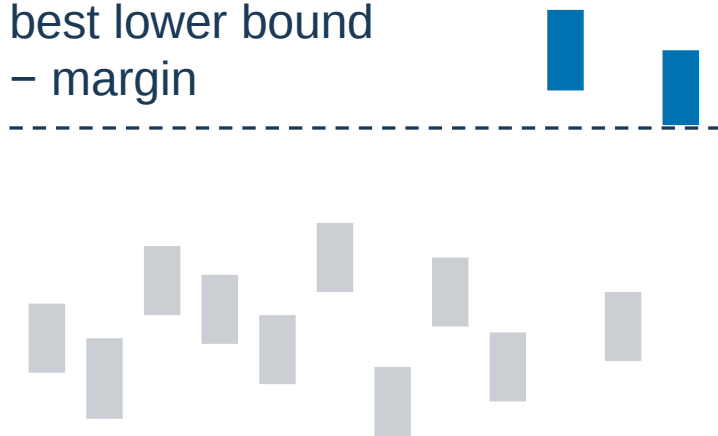
Never from the head GEMM or its tie rule

**So every comparison fixes the batch shape and configuration**

# Bound every score, recompute the few that could win

## 1 Screen with int8

best lower bound  
– margin



entries below the line  
cannot win: dropped

## 2 Recompute a few

a



b



exact FP64 scores,  
widened by the stock  
kernel's error bound

## 3 Decide or fall back

one token under every  
admissible accumulation:  
return it

still open:  
run the stock head

Schematic. The int8 copy reads half the head's bytes; the stock kernel's error bound is an assumed, tested model.

# About 1.8 recomputations per decision, every token equal

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**1.8**

entries recomputed per decision

median 1; kernel replay of 60,000 positions

**0.28–1.46%**

of positions left to the stock head

depending on the error model

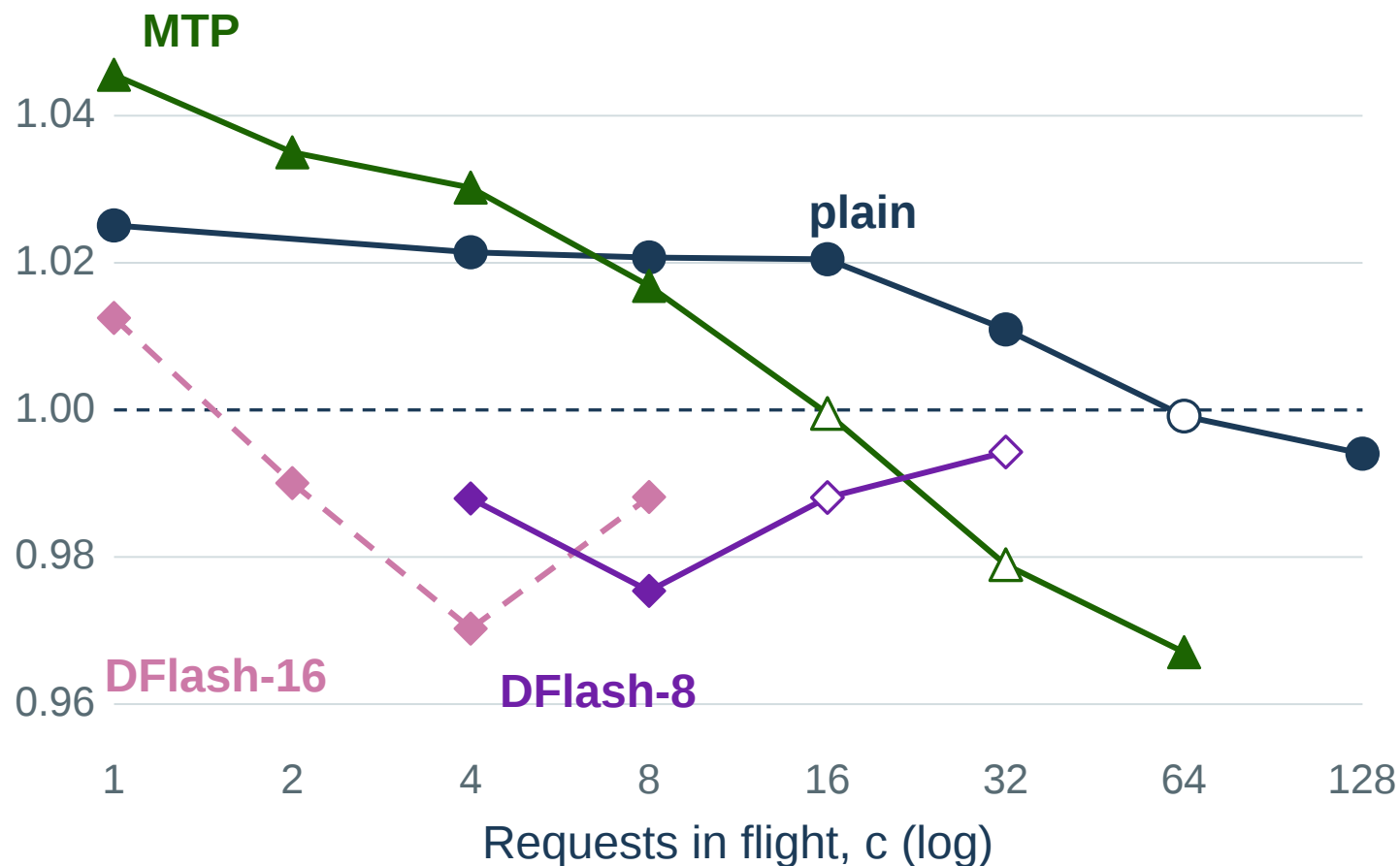
**0 of 592,433**

served positions differ from the stock head

check mode runs the stock head beside the certified one; every certified call counted

# The head gains 1–5% at $c = 1$ , slows the fastest arm at 2–8

Throughput with the head  $\div$  without



**+1.2–4.6%**

at  $c = 1$ , 28–71% of the predicted gain

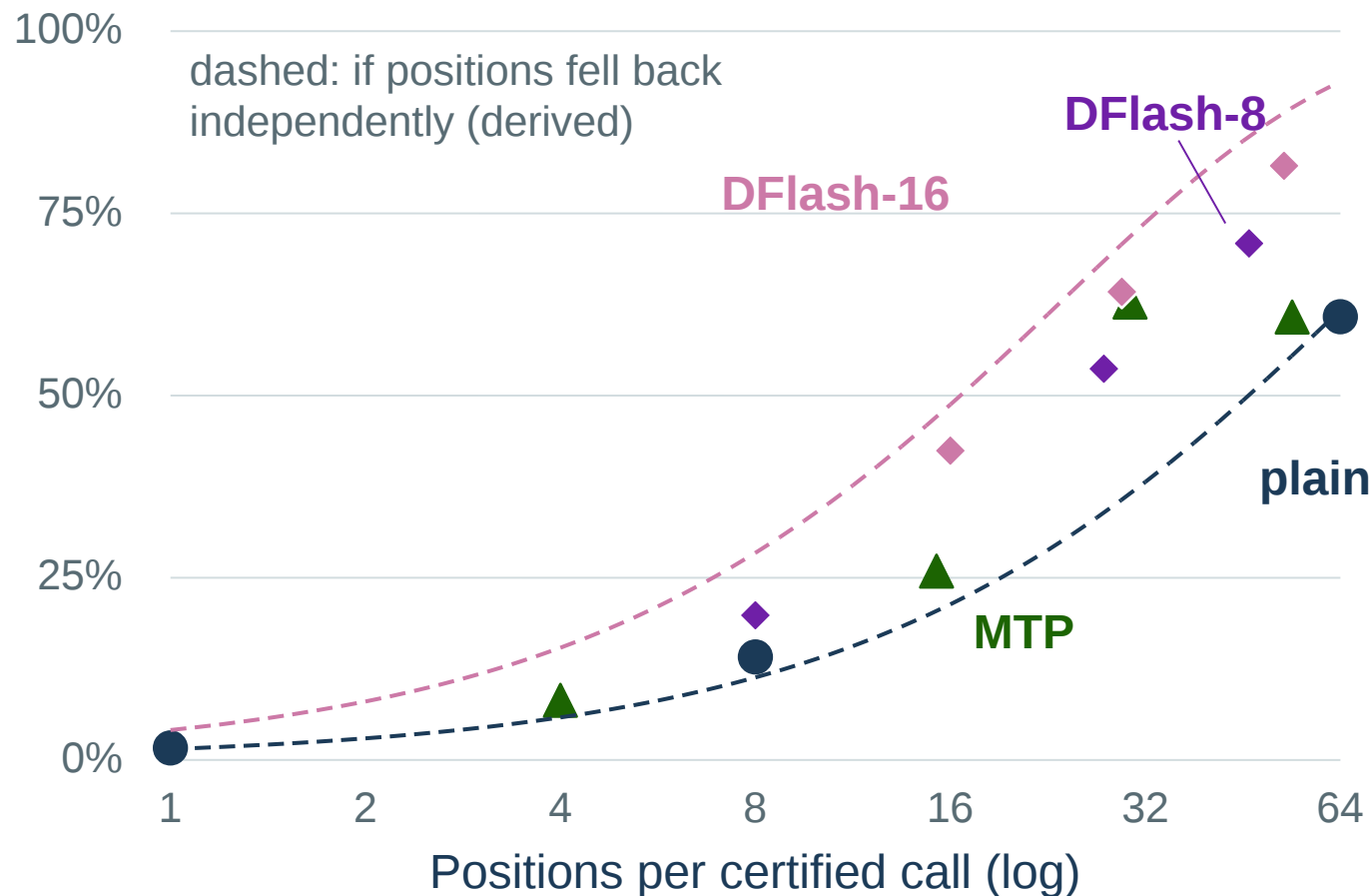
**–1.0–3.0%**

for the fastest arm at  $c = 2–8$

Filled: 95% interval excludes 1.  
Three sessions, each arm against itself.

# Most DFlash verify calls above $c = 1$ need a fallback

Calls with a fallback (check mode)



**54–82%**

of DFlash verify calls at  $c = 2–8$  need a fallback

**0.65×**

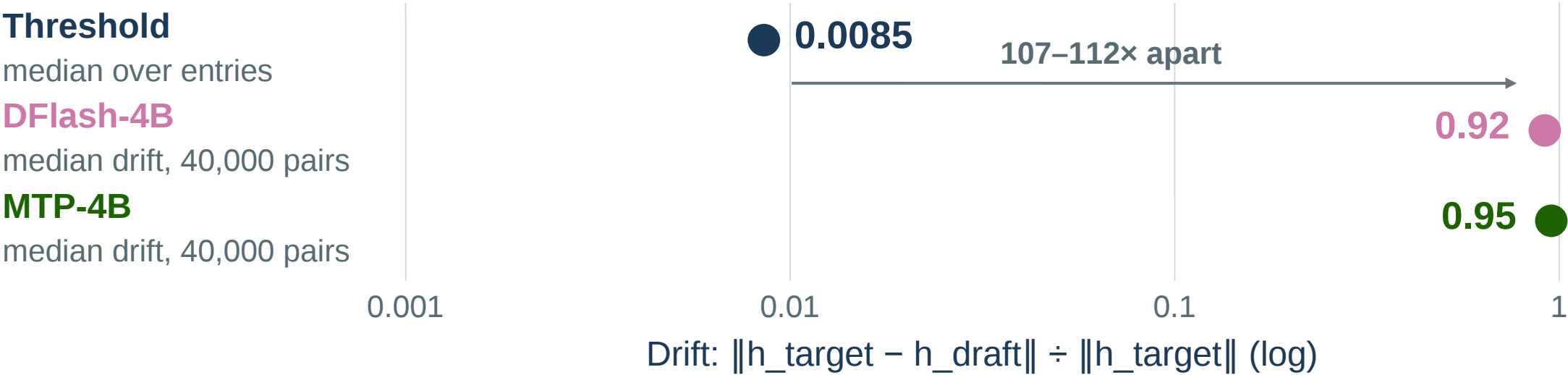
the stock head's time for a fully certified call at one position: the int8 pass is not memory-bound

Fallback cost inside the served graph: not timed

# Transport needs drift below 0.01; median drift is above 0.9



Made-up numbers: the drafter’s scores, carried to the target’s head input by a Cauchy–Schwarz bound



**Certified transport skips at most 0.59% (DFlash-4B) and 0.64% (MTP-4B) of the vocabulary.**

# Engine changes outside the head gained more

fold: SGLang’s fold-every-commit GDN verify, without per-position state snapshots · FA4: FlashAttention 4

## Certified head

three arms · c = 1 · 3 sessions



## Replayed GDN decoding

plain · c = 128 · one run pair



## Fold, narrow tiles, FA4

DFlash-16 · c = 1–4 · per request · 3 sessions



## Fold, FA4 target

DFlash-8 · c = 8–32 · 3 sessions



## Prefill delay and cap

MTP · c = 48–128 · 3 sessions



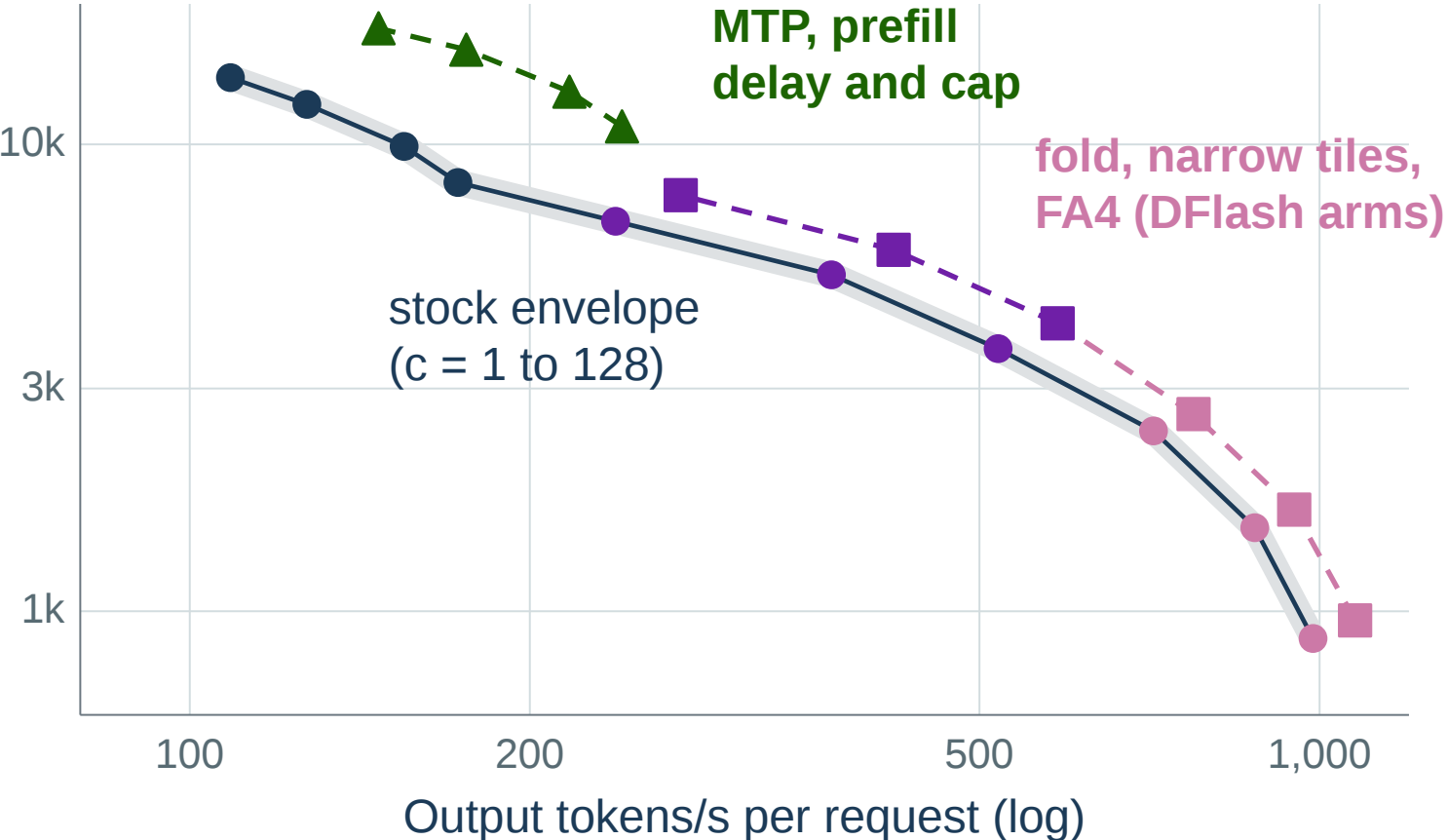
Light: up to the smallest gain; dark: the range across c. Each against the tuned arm it changes; below the line, against the best arm without speculation.

# Six approaches fell short, each by a measured margin

| Approach               | Measured                                                                  | Scope          |
|------------------------|---------------------------------------------------------------------------|----------------|
| Transport              | median drift above 0.9; it beats int8 only below about 0.01               | 2 drafters     |
| FP8 dense layers       | per-row arm 1.014× at c = 64 (rule 1.05); faster arms not shown in budget | 1 session each |
| Int4 target + drafter  | probe top-1 0.969–0.975 (floor 0.98); 0.71–0.87× above c = 1              | 3 sessions     |
| FP16 recurrent state   | 1.16–1.17× at c = 64–256; 1.0–1.4 GSM8K points below (budget 1)           | 3 sessions     |
| Fold + routing + head  | below its declared range at c = 2–8 (0.972× at 2 vs ≥ 1.007)              | 3 sessions     |
| Split-KV verify kernel | saved 7.6 of the 200 μs its rule required                                 | microbenchmark |

# Both confirmed changes plot beyond the stock envelope

Output tokens/s on the GPU (log)



**+8–14%**

at c = 1–32 against the  
same DFlash arms

**1.18–1.32×**

the best plain arm at  
c = 48–128, first tokens  
later

Ratios within each  
confirmation's sessions; the plot  
combines sessions.

# Where compared, outputs match stock up to rounding

Greedy outputs against stock, same configuration

## Certified head

identical: 0 of 592,433 checked positions; tokens at  $c = 1$

## Fold verify

bitwise, with matched pools and batches

## Fold, narrow tiles, FA4

up to rounding, classed at  $c = 1$

## Prefill delay and cap

up to rounding, classed at  $c = 64$  and  $128$

Up to rounding: each first divergence from a matched stock reference is a tie, one BF16 step or a near tie.

GSM8K, 1,319 problems, sampled

**-1.36 to +1.06**

points for four arms against two plain runs, which differ by  $+0.30$

**Within plain decoding's noise: it rules out losses above about 3.3 points, not smaller ones**

# Hold the fast path to the engine's own token

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The design, a fast path held to the engine's output with its kernel as fallback, is not specific to the head.

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## Limits

The error model is assumed and tested on one GPU, not proved

Fallback cost inside the served graph was not timed

One model, engine commit, GPU and workload

## Next

Time the in-graph fallbacks

Train a drafter for certificate-friendly states

A delay-only admission arm

Batched equality at  $c > 1$  for the fold with FA4

Natural output lengths at  $c = 32$  and  $128$

# Backup slides, by question

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|                                                                  |         |
|------------------------------------------------------------------|---------|
| <b>Is it really exact? The contract and the error model</b>      | B02–B09 |
| <b>Why is the served gain so small?</b>                          | B10–B14 |
| <b>Why does transport fail?</b>                                  | B15–B18 |
| <b>The engine changes and their exactness classes</b>            | B19–B24 |
| <b>The profile in detail, related work, upstream fixes, code</b> | B25–B29 |

# Certify the stock kernel's token, or run the stock kernel

Bound each logit's FP32 sum, round the bounds to BF16, then:

**Every admissible sum gives the same first maximum** → **Return that token**

**The bounds leave the token open** → **Run the stock head**

**Whole-batch fallback:** the stock matmul at the same M; no error model

**Per-position fallback:** the open position's candidates; needs batch-invariant logits (held at 20 batch sizes)

## The error model

**Conservative:** 2D roundings of  $2^{-23}$ , any order

**Hopper:** tensor-core blocks of 16, 25 bits, truncation, plus split-K

**Tested:** 14 cuBLAS kernels on crafted inputs; FP32-output error 1/58 of the bound

**Not covered:** BF16 split-K partial sums

# The reference is the engine's actual result

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| Reference | Computation                        |
|-----------|------------------------------------|
| R-real    | Exact dot products                 |
| R-bf16    | Exact dot products → BF16 rounding |
| R-stock   | Existing kernel → BF16 rounding    |

Same inputs, batch shape and configuration.

Assumed error models are not vendor guarantees.

# Fixed ties certify; uncertain ties fall back

| Case          | Token 17 | Token 42         | Action    |
|---------------|----------|------------------|-----------|
| Separated     | 1.0      | 0.99609375       | Return 17 |
| Fixed tie     | 1.0      | 1.0              | Return 17 |
| Uncertain tie | 1.0      | 1.0 or 1.0078125 | Fallback  |

Rounded-score bounds for a candidate k:

Earlier j:  $\text{upper}(j) < \text{lower}(k)$       Later j:  $\text{upper}(j) \leq \text{lower}(k)$

# Check mode found no differing token in 592,433 positions

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**0 of 592,433**

certified positions differ from the stock head run beside it

**1.03–1.21×**

closed-loop divergence, certified vs stock, over stock vs stock

**3.8–25.7 GB**

more memory for the certified CUDA graphs

**0.6–1.2%**

cost when gated off, at calls above 64 positions

# Every exact change matches stock up to rounding-level ties

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|                           |                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Same configuration</b> | Host-gap patches: tokens identical on 192 of 192 requests<br>Fold verify (matched pools): bitwise; with FA4: ties, one BF16 step or near ties ( $c = 1$ )<br>Certified head: 0 of 592,433 positions differ in check mode |
| <b>Closed loop</b>        | Batches evolve differently run to run: certified vs stock diverges at 1.03–1.21× the stock-vs-stock rate                                                                                                                 |
| <b>Task accuracy</b>      | GSM8K, sampled: exact arms within −1.36 to +1.06 points of plain decoding; it cannot exclude losses under about 3.3 points                                                                                               |

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# SGLang’s typical BF16 error matches Hugging Face’s

Mean |log-probability error| on FP32’s top token (nats); misses above 2 nats

| Implementation, path                     | Sample 1: 6,144 | Sample 2: 9,216 | Misses > 2 nats |
|------------------------------------------|-----------------|-----------------|-----------------|
| <b>SGLang (pin, unpatched)</b> · decode  | 0.0054          | 0.0039          | 0               |
| <b>SGLang (pin, unpatched)</b> · prefill | 0.0054          | 0.0045          | 1               |
| <b>transformers, torch GDN</b> · decode  | 0.0057          | 0.0044          | 0               |
| <b>transformers, torch GDN</b> · prefill | 0.0056          | 0.0049          | 1               |
| <b>transformers, fla GDN</b> · decode    | 0.0056          | 0.0042          | 0               |
| <b>transformers, fla GDN</b> · prefill   | 0.0056          | 0.0047          | 1               |

The 8.9-nat miss is an SGLang outlier at one position; a second position is ill-conditioned: FP32’s own top token flips under rounding-size noise. No issue filed.

# Quality results do not establish equivalence

| GSM8K · 1,319 problems  | Accuracy        |
|-------------------------|-----------------|
| Plain run 1 / run 2     | 89.99% / 90.30% |
| Buffered MTP            | 89.54%          |
| MTP, snapshot verify    | 88.93%          |
| DFlash, block 8         | 89.69%          |
| Buffered plain decoding | 91.05%          |

**Paired 95% intervals allow  $\approx$ 3.3 percentage points of loss.**

# Seeded sampling needs fixed noise per token

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$z_i$ : score ·  $T > 0$ : temperature ·  $G_i$ : Gumbel noise

**Choose the largest  $z_i / T + G_i$ .**

Noise key: seed, sequence index, token ID

Uncertain choice → stock sampler

With its fallback it matches SGLang's seeded sampler: 60,000 replayed positions at  $T = 0.7$  and 1, and 21,016 in SGLang at  $T = 0.7$

Acceptance against the stock rejection sampler: not measured

# Alone, the int8 head takes about 0.7 of the stock head's time

Head alone, one call of M positions

One position, measured

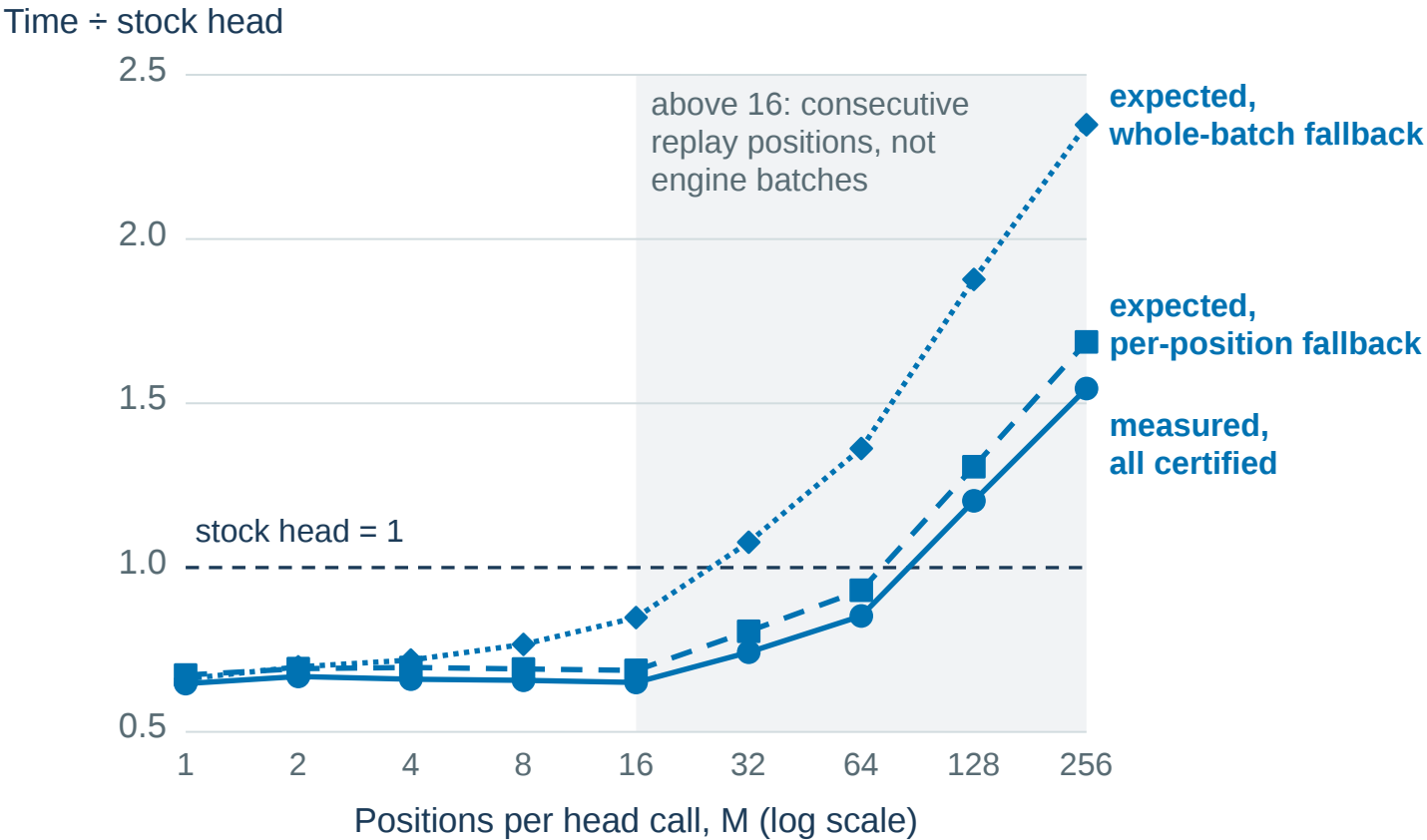
370.5  $\mu$ s

stock head: GEMM, FP32 copy, argmax

239.6  $\mu$ s

certified head, every position certified

With the head at 10.3% of a batch-1 plain step, a step would shorten by at most 3.4–3.5% (derived); served, plain decoding gained 2.5%.



# Int8 leaves few candidates; fallback grows with the batch

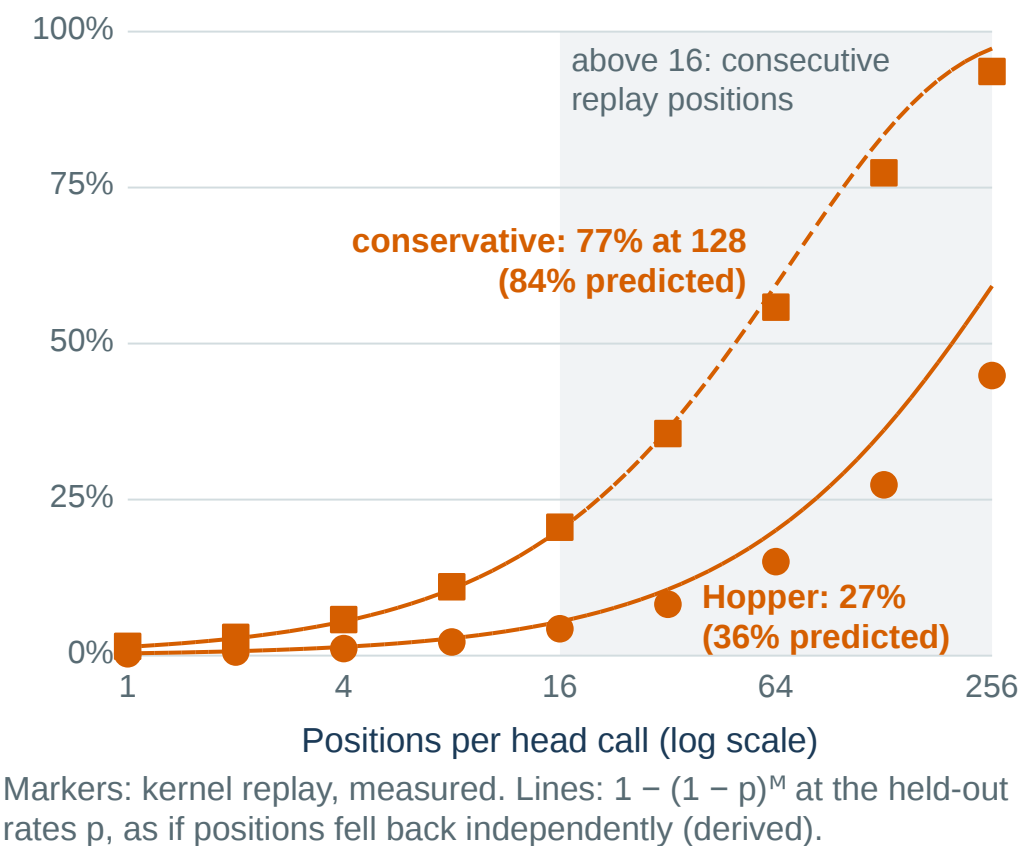
Candidates per greedy decision, by copy of the head (held-out replay, 6,005 positions, exact reference)

| Copy                | Bytes | Mean   | p99     | Max     |
|---------------------|-------|--------|---------|---------|
| Int8, per entry     | 0.50  | 1.55   | 8       | 30      |
| Int8, 128 columns   | 0.52  | 1.34   | 5       | 23      |
| Int8, 32 columns    | 0.54  | 1.26   | 5       | 20      |
| FP8 E4M3, per entry | 0.51  | 4.89   | 60      | 605     |
| Int4, 128 columns   | 0.27  | 72,847 | 248,295 | 248,320 |

**Why int8 is enough:** its intervals are about two logits wide; the median gap between the two largest logits is 7.3 logits.

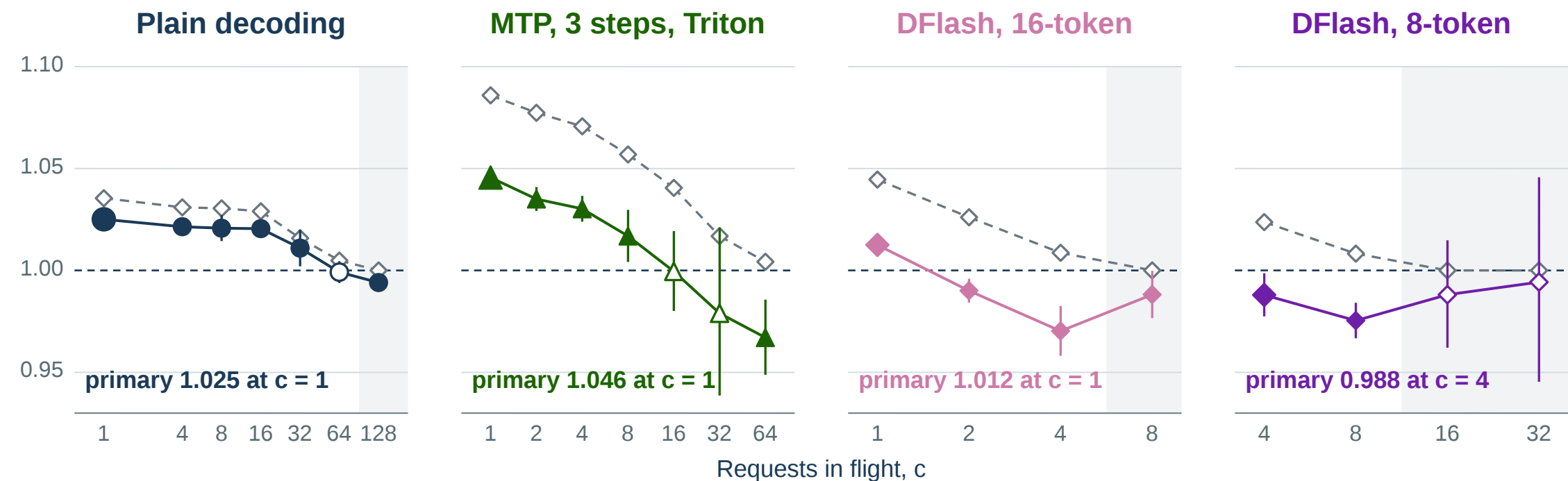
**Held-out fallback:** 0.35% (Hopper model) and 1.40% (conservative) of positions under the interval condition; the gap condition never goes below about 2%.

Calls with at least one open position, which the whole-batch fallback sends to the stock head



# Every served arm fell short of its prediction

Throughput with the certified head ÷ without, each tuned arm against itself



Filled: 95% interval excludes 1 · bars: 95% intervals · grey ◇: prediction · shaded: head gated off

At c = 1 the gain is 71% (plain), 53% (MTP) and 28% (16-token DFlash) of the prediction, tokens identical; gains continue to c = 32 (plain) and 8 (MTP); both DFlash arms lose 1.0–3.0% where the head is active.

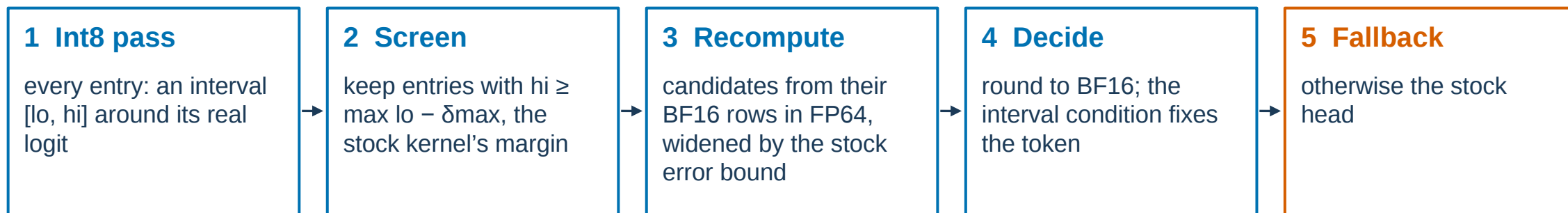
# Certified head served: every point against its prediction

Throughput ratio, certified ÷ stock; second line: predicted before the runs

| Arm              | c = 1                        | c = 2                        | c = 4                        | c = 8                        | c = 16                | c = 32                | c = 64                       | c = 128                      |
|------------------|------------------------------|------------------------------|------------------------------|------------------------------|-----------------------|-----------------------|------------------------------|------------------------------|
| Plain decoding   | <b>1.025</b><br><i>1.035</i> |                              | 1.021<br><i>1.031</i>        | 1.021<br><i>1.030</i>        | 1.020<br><i>1.029</i> | 1.011<br><i>1.016</i> | 0.999<br><i>1.005</i>        | <b>0.994</b><br><i>1.000</i> |
| MTP, 3 steps     | <b>1.046</b><br><i>1.086</i> | 1.035<br><i>1.077</i>        | 1.030<br><i>1.071</i>        | 1.017<br><i>1.057</i>        | 1.000<br><i>1.041</i> | 0.979<br><i>1.017</i> | <b>0.967</b><br><i>1.004</i> |                              |
| DFlash, 16-token | <b>1.012</b><br><i>1.045</i> | <b>0.990</b><br><i>1.026</i> | <b>0.970</b><br><i>1.009</i> | <b>0.988</b><br><i>1.000</i> |                       |                       |                              |                              |
| DFlash, 8-token  |                              |                              | <b>0.988</b><br><i>1.024</i> | <b>0.975</b><br><i>1.008</i> | 0.988<br><i>1.000</i> | 0.994<br><i>1.000</i> |                              |                              |

The predictions assumed plain decoding’s fallback rate on every path; drafted positions fall back more.

# Int8 screens every entry; a few are recomputed exactly



## The error bound of the copy

$$\beta = \|e\| \|h\| + \gamma \|\bar{w}\| \|h\| + \eta'$$

e: the copy's known error per entry;  $\gamma$ : tensor-core accumulation (conservative model).

The term  $\|e\| \|h\|$  assumes the worst alignment of e and h. Its median on captured states is 1.00 logit, about 75 times the realized error (0.013).

The copy is 0.64 GB beside the BF16 head; one pass reads half the head's bytes.

## The kernel, and seeded sampling

A W8A16 GEMM computes each interval in its epilogue with directed rounding and keeps, per chunk, the four largest upper ends; one CUDA graph per batch size holds it and the fallback.

Each compiled variant self-tests before it certifies; every call checks eight exact entries per position. The checks caught two faults, both refused.

The same steps on Gumbel-noised scores certify SGLang's seeded sampler's token: 60,000 replayed positions at  $T = 0.7$  and 1, 21,016 in SGLang at  $T = 0.7$ .

# Drafts agree with the target on tokens, not on head inputs

## 50.6 vs 159

DFlash-4B's head input norm is always  $\sqrt{2560}$  (unit-gain final RMSNorm), a third of the target's median

Norms alone give drift  $\geq 0.68$

## 0.91

drift left after the best rescaling of the draft input (median cosine 0.41)

The angle is large too

## 0.91 / 0.99

median drift where the draft token was accepted (DFlash-4B / MTP-4B)

Same token, different head input

**Per position, draft and target logits differ by 2.0–2.1 RMS (medians):** draft scores are not small perturbations of target scores.

**The int8 copy leaves about 1.6 candidates** on the DFlash-4B verification positions a greedy verifier needs.

**What would rescue it:** a drafter whose head input tracks the target's. Scaling the measured drift down 10× would let the same bounds skip 49% (DFlash-4B) of one position's vocabulary; 100×, 84% (a model extrapolation). Batches would still need most of it: even oracle radii need 97% for random groups of 16 DFlash-4B pairs.

# Transport's uncertainty is radius × change in head input

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## Tile geometry

$w_i$ : entry's weight vector

$\mu$ : centre;  $r$ : radius

$$\|w_i - \mu\| \leq r$$

## State change

$\Delta$  = target – draft

$M$ : draft tile maximum

$$\text{Target maximum} \leq M + \mu \cdot \Delta + r\|\Delta\|$$

Skip only when this bound is below a known target score.

# Changing the tiles did not rescue transport

Vocabulary excluded · T: transport · S: static bound without draft scores

| Entries / tile   | DFlash T | DFlash S | MTP T | MTP S |
|------------------|----------|----------|-------|-------|
| Contiguous 64    | 0.08%    | 0.08%    | 0.08% | 0.08% |
| Contiguous 256   | 0.00%    | 0.00%    | 0.00% | 0.00% |
| Contiguous 1,024 | 0.00%    | 0.00%    | 0.00% | 0.00% |
| Random 256       | 0.00%    | 0.00%    | 0.00% | 0.00% |
| k-means 64       | 0.59%    | 0.58%    | 0.64% | 0.64% |
| k-means 256      | 0.10%    | 0.10%    | 0.11% | 0.11% |
| k-means 1,024    | 0.02%    | 0.02%    | 0.02% | 0.02% |

DFlash: 4,020 pairs · MTP: 16,016 pairs

# Transport leaves over 99% of vocabulary entries to check

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**DFlash**

**99.4% remain**



**MTP**

**99.4% remain**



**Little benefit beyond bounds without draft scores.**

Best tiles on the two tested drafters

# Every change against the arm it modifies

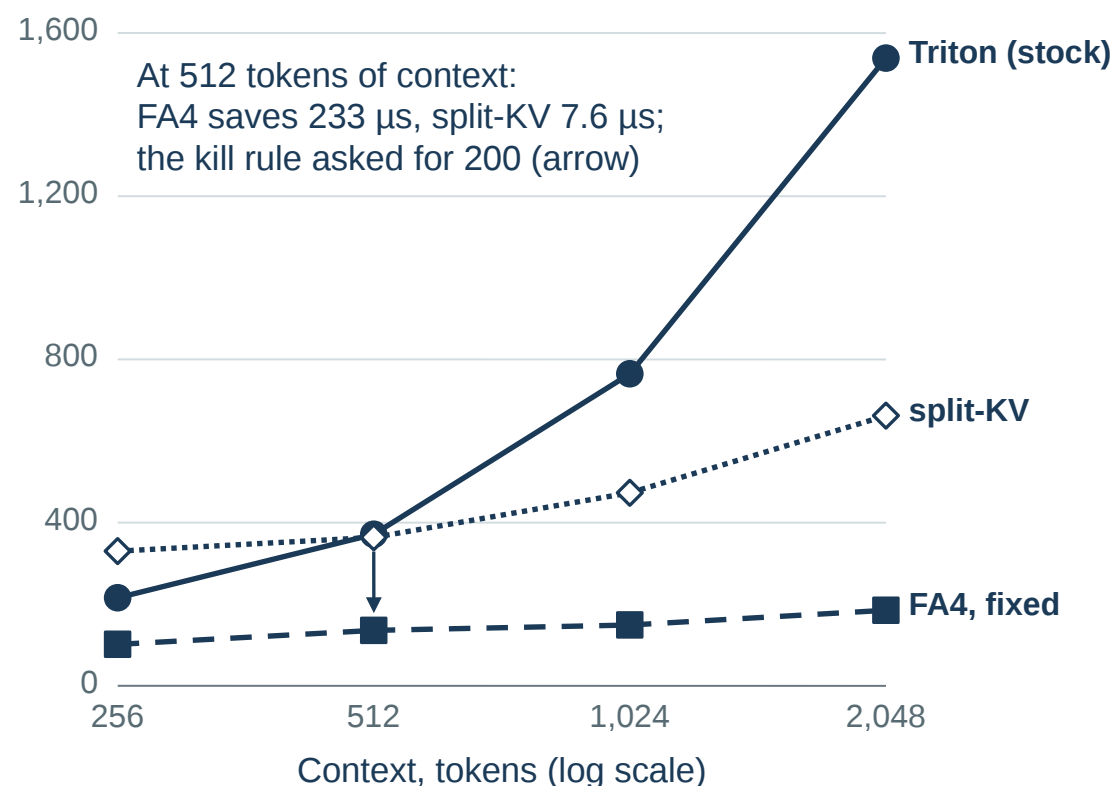
Ratio with ÷ without; brackets: run-pair range, session range or 95% interval

| Change                  | Arm                | c         | Ratio                       | Sess. | Outputs            |
|-------------------------|--------------------|-----------|-----------------------------|-------|--------------------|
| Host-gap patches        | MTP                | 1–8       | 1.094–1.096 (1.090–1.101)   | 1     | token-identical    |
| Prefill delay, cap 16   | MTP                | 32–128    | 1.19–1.48 (1.17–1.50)       | 3     | rounding (64, 128) |
| Backbone GEMM routing   | plain              | 1         | 1.035 (1.034–1.035)         | 1     | rounding           |
| Replayed GDN decoding   | plain              | 128       | 1.081 (one pair)            | 1     | rounding           |
| Fold verify             | DFlash-16          | 1 · 8–32  | 0.968 · 1.061–1.120         | 1     | bitwise            |
| Fold, narrow tiles, FA4 | DFlash-16          | 1–4       | x 1.084–1.093 (1.056–1.114) | 3     | rounding (c = 1)   |
| Fold, FA4 target        | DFlash-8           | 8–32      | 1.124–1.137 (1.108–1.166)   | 3     | rounding (c = 1)   |
| Certified head          | plain · MTP · D-16 | 1         | 1.025 · 1.046 · 1.012       | 3     | identical          |
| Certified head          | D-16 · D-8         | 2–4 · 4–8 | 0.970–0.990 · 0.975–0.988   | 3     | check mode         |

“rounding”: exact up to rounding, as classed at the c given. The prefill delay against the best non-speculative arm: 1.18–1.32× at c = 48–128.

# The fold, narrow tiles and FA4: 8–14% at c = 1–32

Target attention, 8 layers, one request:  $\mu\text{s}$  per forward (microbenchmark)



## 80 // 128 = 0

Qwen3.5's head-dim-256 tile on sm\_90 is  $128 \times 80$ : floor division gave 128 loader threads no page-table entries, so FA4 never compiled. The fix is in flash-attention and in SGLang PR #35757 (open); we backported it.

## 1.084–1.093

per request, 16-token DFlash, c = 1–4: fold, narrow tiles, FA4 for drafter and target

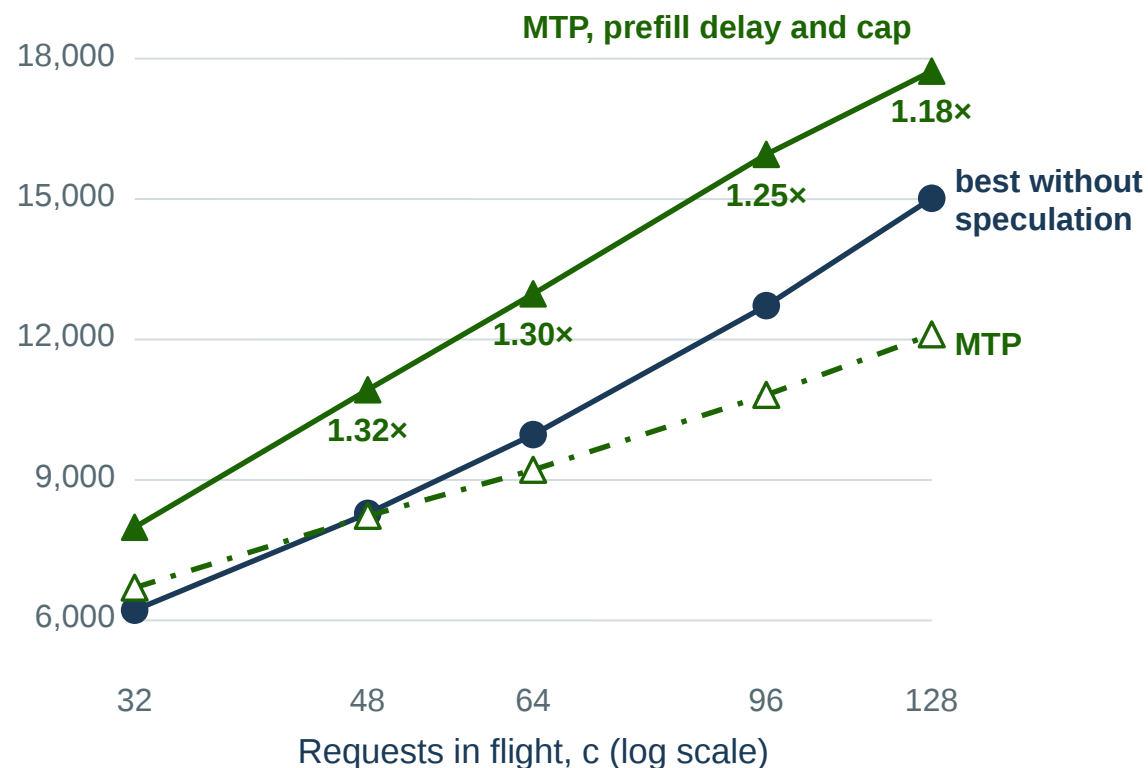
## 1.124–1.137

throughput, 8-token DFlash (already drafting with FA4), c = 8–32: fold, FA4 target

Exact up to rounding as classed at c = 1 (the fold alone bitwise); batched outputs not compared. Alone at c = 1–4 each FA4 change is a speedup (drafting 1.025–1.032, target 1.030–1.038 per request); at 4 neither is shown to shorten the cycle. The fold alone at 4: 1.005 (0.990–1.021).

# A capped prefill delay puts MTP ahead at $c = 48\text{--}128$

Throughput, mean of three sessions; labels:  $\times$  the best arm without speculation



## 33 ms

per lone prefill, 9 ms of it GPU work. Speculative requests finish one or two at a time and are prefilled alone; plain decoding's equal-length requests finish together and share a prefill.

## 1.18–1.32x

the best arm without speculation, every session; prefill batches at  $c = 128$ : 518  $\rightarrow$  81

## 2.6–3.1x

plain decoding's 99th-percentile time to first token, at  $c = 48\text{--}128$  (385–635 ms)

Exact up to rounding against MTP without it, classed at  $c = 64$  and 128. At  $c = 1$  and 8 it passes the no-harm rule (0.994, 0.992), but median time to first token rises 0.6–2.7 ms in every session pair. One setting, not tuned; no arm separates the delay from the cap.

# Both declared approximations miss the quality budget

Budget: GSM8K within 1.0 point; logit probe top-1  $\geq 0.98$ , KL  $\leq 0.01$

| Approximation           | Throughput vs best non-lossy | Quality                        | In budget |
|-------------------------|------------------------------|--------------------------------|-----------|
| FP16 recurrent state    | 1.16–1.17× at c = 64–256     | GSM8K –1.0 to –1.4 points      | No        |
| Int4 target and drafter | 0.71–0.87× above c = 1       | probe fails: top-1 0.969–0.975 | No        |

FP16 state halves the recurrent state; its GSM8K loss is within the exact arms’ own spread (–1.36 to +1.06), which this check cannot resolve.

Int4: the W4A16 GEMM is slower than BF16 from 32 rows per call, and the int4 drafter accepts fewer tokens.

# The first composition fell below its declared range

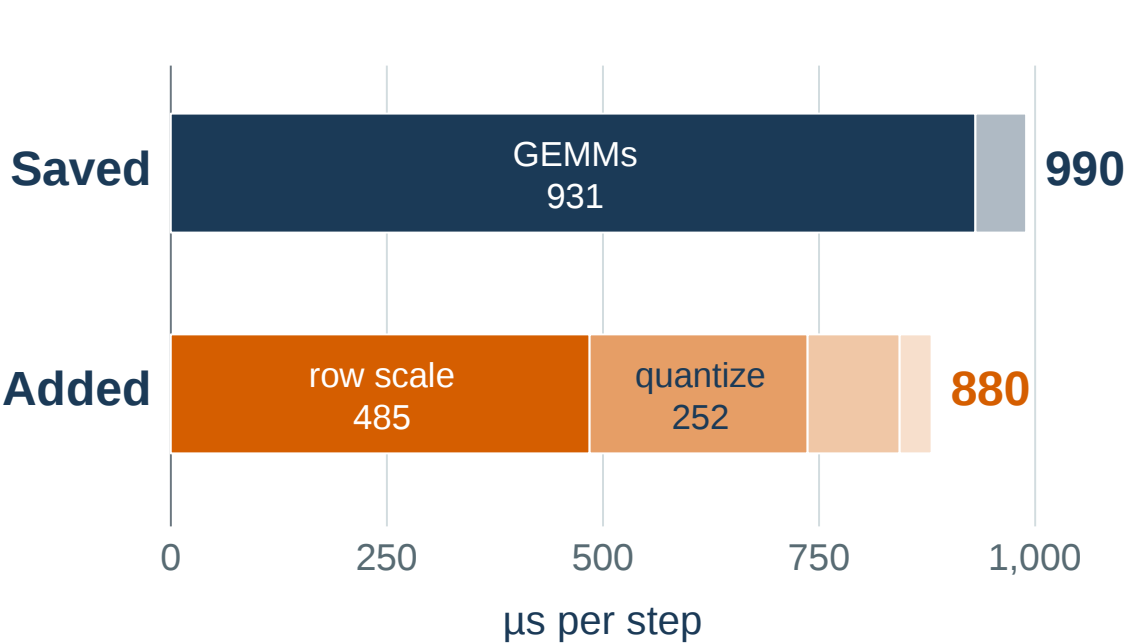
Against tuned 16-token DFlash: ratio and 95% interval

| Arm                             | c = 1                | c = 2                | c = 4                | c = 8                |
|---------------------------------|----------------------|----------------------|----------------------|----------------------|
| Fold + routing + certified head | 0.986<br>0.982–0.989 | 0.972<br>0.966–0.979 | 1.013<br>1.007–1.020 | 1.079<br>1.063–1.095 |
| Fold verify                     | 0.970<br>0.967–0.974 | 0.987<br>0.984–0.989 | 1.009<br>1.002–1.016 | 1.073<br>1.065–1.081 |
| Routing table                   | 1.009<br>1.004–1.014 | 0.988<br>0.985–0.991 | 1.017<br>1.011–1.022 | 1.008<br>0.995–1.021 |
| Fold + routing                  | 0.974<br>0.967–0.982 | 0.971<br>0.959–0.984 | 1.021<br>1.008–1.033 | 1.082<br>1.058–1.107 |
| Composed tree, switches off     | 0.999<br>0.996–1.003 | 0.998<br>0.995–1.002 | 0.999<br>0.990–1.009 | 0.995<br>0.983–1.007 |

The later composition, the fold with narrow tiles and FA4, gained +9.3% per request at c = 1.

# FP8 halved the weight bytes; added kernels took it back

Plain decoding, one request:  $\mu$ s per decode step, FP8 with per-row scales against BF16 (Nsight)



**1.014x**

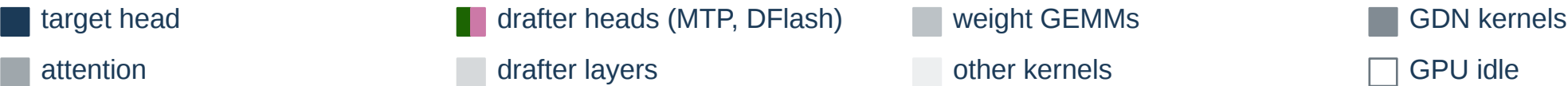
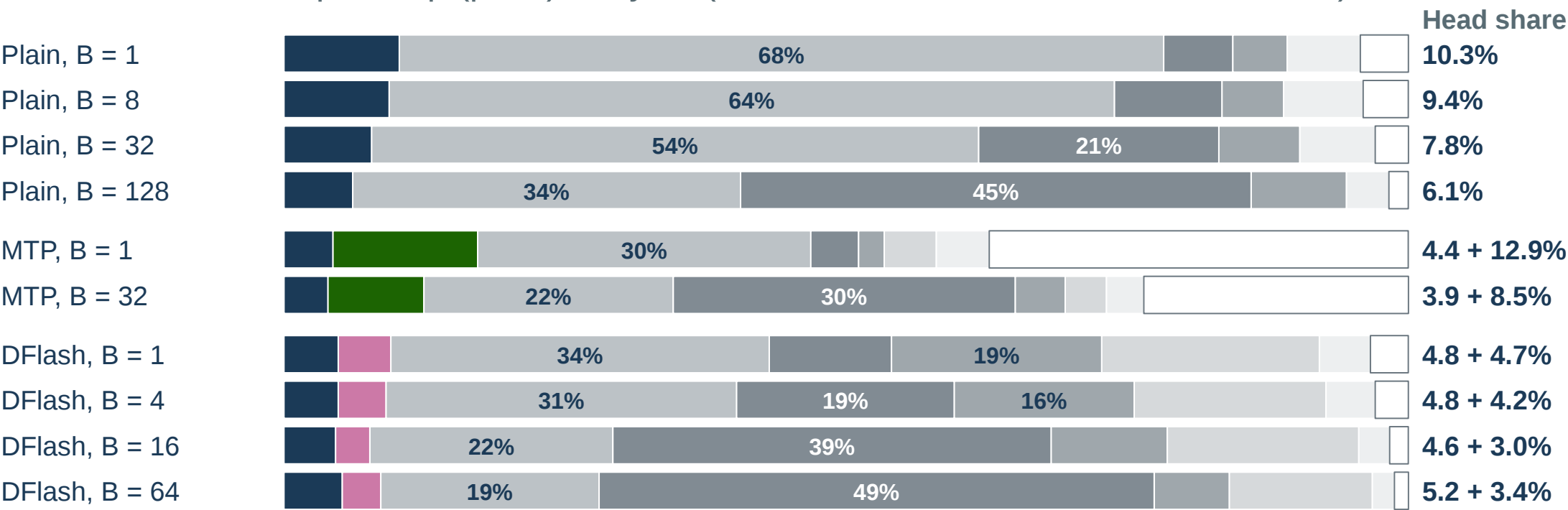
decode rate at  $c = 64$  against a kill rule of 1.05

**1.120–1.260x**

without the row-scale kernel, at  $c = 1\text{--}64$ ;  
neither variant shown inside the quality budget

# Weights dominate at batch 1, recurrent state at batch 128

Share of GPU time per step (plain) or cycle (MTP, untuned; 16-token DFlash, tuned)



# New here: certifying a closed-source kernel's token

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## Prior art

Certified low-precision heads: Sparkpipe, dgpp, a Metal engine that replicates its kernel, knlp

Bound-and-refine search; cluster bounds (the static screen)

Sampling from bounded scores; drafter-invariant verification

## This paper

A contract with a closed-source kernel's rounding, under a tested error model, with the stock head as fallback

Transport of draft-side tile summaries: no precedent found, and a negative result

# Seven pull requests and six reports upstream, all open

| Item                                   | What it fixes or reports                                                                                                                                            | Status                            |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>flash-attention #2957, #2958</b>    | FA4's SM90 paged-KV loader copies rows past a partial tile (head dims 80, 96, 160, 224); the PR skips those copies                                                  | Open, no reply<br>Open, no review |
| <b>Comment on SGLang #35757</b>        | Confirms the head-dim-256 compile fix on Qwen3.5; reports the loader bug with the vendored-copy diff                                                                | Open, no reply                    |
| <b>Comment on SGLang #42019</b>        | max(1, floor) left tile_n 144 short. The author adopted the ceiling division; on an H200 it alone does not fix tile_n 144. We pointed to #2958 and offered the diff | Open; author replied, we answered |
| <b>SGLang #42263</b>                   | Builds sm_90a for the aarch64 sgl-kernel wheel, whose SM90 CUTLASS GEMMs computed nothing on GH200                                                                  | Open, no review                   |
| <b>SGLang #42209, #42195</b>           | The fold for DFlash and MTP; FlashInfer planning without blocking reads (both opt-in)                                                                               | Open, no review                   |
| <b>SGLang #42160, #42126, #42094</b>   | 64-bit offsets in a fused kernel; the packed GDN projection on CUDA; a tied MTP drafter start-up crash                                                              | Open, no review                   |
| <b>Issue #42085; on #41351, #38118</b> | GEMV flag ignored by unquantized layers; radix-cache history changes log-probabilities; a chunked tail counted twice                                                | Open, no maintainer reply         |

# One repository holds the code, harness and evidence

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## **src/certified\_head/**

the int8 head: Triton kernels, bounds, self-tests

## **engine/sglang/patches/**

opt-in SGLang patch series, one per topic

## **bench/**

serving harness: arms, launch checks, sweeps, Pareto

## **experiments/, evidence/**

scripts; committed results, each README with its command

## **tests/, formal/**

exact CPU references, pytest, Lean 4 lemmas

# CPU checks are separate from GPU campaigns

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[github.com/SamMausberg/verified-progress](https://github.com/SamMausberg/verified-progress)

```
python -m venv .venv
. .venv/bin/activate
python -m pip install -r requirements-cpu.txt \
    -r requirements-dev.txt
python scripts/verify_artifact.py
python tests/test_precision.py
python tests/test_contracts.py
python tests/test_state_structure.py
python -m pytest tests/
```

GPU procedures and pinned environment: see speaker notes.